

2.2. Examples

The following examples show the relevance of FDR control in some typical situations. In addition they indicate the desirability of a large number of rejections (discoveries). Both aspects are addressed by the procedure given later in Section 3.

One type of multiple-comparison problem involves an overall decision (conclusion, recommendation, etc.) which is based on multiple inferences. An example of this type of problems is the 'multiple end points problem', which was used earlier to show that FWER control is not always needed. In this example the overall decision problem is whether to recommend a new treatment over a standard treatment. Discoveries here are rejections of null hypotheses claiming that treatment is no better than standard on specified end points. These conclusions about different aspects of the benefit of the new treatment are of interest *per se*, but the set of discoveries will be used to reach an overall decision regarding the new treatment. We wish therefore to make as many discoveries as possible (which will enhance a decision in favour of the new treatment), subject to control of the FDR. Control of the probability of any error is unnecessarily stringent, as a small proportion of errors will not change the overall validity of the conclusion.

Another type of problems involves multiple separate decisions without an overall decision being required. An example of this type is the multiple-subgroups problem, where two treatments are compared in multiple subgroups, and separate recommendations on the preferred treatments must be made for all subgroups. As usual we wish to discover as many as possible significant differences, thereby reaching operational decisions, but would be willing to admit a prespecified proportion of misses, i.e. willing to use an FDR controlling procedure.

The third type involves screening problems, where multiple potential effects are screened to weed out the null effects. One example is screening of various chemicals for potential drug development. Another example is testing multiple factors in an experimental (2^k say) design. In such examples we want to obtain as many as possible discoveries (candidates for drug developments, factors that affect the quality of a product) but again wish to control the FDR, because too large a fraction of false leads would burden the second phase of the confirmatory analysis.

2.3. Alternative Formulations

We have suggested to capture the error rate vaguely described as 'the proportion of false discoveries' using the FDR. At this point it might be illuminating to discuss alternative formulations of this concept, and thus to motivate our choice of FDR further.

Undoubtedly, controlling the random variable $\mathbf{Q}$ at each realization is most desirable. This is impossible, for if $m_0 = m$ and even if a single hypothesis is rejected $v/r = 1$ and $\mathbf{Q}$ cannot then be controlled. Controlling $(\mathbf{V}/\mathbf{R} | \mathbf{R} > 0)$ has the same problem—it is identically 1 in the above configuration. Therefore $E(\mathbf{V}/\mathbf{R} | \mathbf{R} > 0)$ cannot be controlled. The FDR, instead, is $P(\mathbf{R} > 0)E(\mathbf{V}/\mathbf{R} | \mathbf{R} > 0)$, and, as will be shown later, this is possible to control.

Second, consider the formulation that Sorić (1989) gave to 'the proportion of false discoveries among the discoveries' as $Q' = E(\mathbf{V})/r$. This quotient is neither the random variable $\mathbf{Q}$ nor its expectation but is a mixture of expectations and realizations. It is not even the conditional expectation of $\mathbf{Q}$, namely $E(\mathbf{Q} | \mathbf{R} = r) = E(\mathbf{V} | \mathbf{R} = r)/r$, which has again the problem of control for $m_0 = m$.

Third, consider $E(\mathbf{V})/E(\mathbf{R})$. When all hypotheses are true it is identically 1, and again impossible to control. A remedy may be given by either adding 1 to the denominator, a somewhat artificial solution, or by changing the denominator to $E(\mathbf{R}|\mathbf{R} > 0)$. Modifying both numerator and denominator in the same way will again run into problems of control when $m_0 = m$.

3. FALSE DISCOVERY RATE CONTROLLING PROCEDURE

3.1. The Procedure

Consider testing $H_1, H_2, \dots, H_m$ based on the corresponding p -values $P_1, P_2, \dots, P_m$. Let $P_{(1)} \leq P_{(2)} \leq \dots \leq P_{(m)}$ be the ordered p -values, and denote by $H_{(i)}$ the null hypothesis corresponding to $P_{(i)}$. Define the following Bonferroni-type multiple-testing procedure:

let k be the largest i for which $P_{(i)} \leq \frac{i}{m} q^*$;

then reject all $H_{(i)}$ $i = 1, 2, \dots, k$. (1)

Theorem 1. For independent test statistics and for any configuration of false null hypotheses, the above procedure controls the FDR at q^* .

Proof. The theorem follows from the following lemma, whose proof is given in Appendix A.

Lemma. For any $0 \leq m_0 \leq m$ independent p -values corresponding to true null hypotheses, and for any values that the $m_1 = m - m_0$ p -values corresponding to the false null hypotheses can take, the multiple-testing procedure defined by procedure (1) above satisfies the inequality

$$E(\mathbf{Q} | P_{m_0+1} = p_1, \dots, P_m = p_{m_1}) \leq \frac{m_0}{m} q^*. \quad (2)$$

Now, suppose that $m_1 = m - m_0$ of the hypotheses are false. Whatever the joint distribution of $P_1'', \dots, P_{m_1}''$ which corresponds to these false hypotheses is, integrating inequality (2) above we obtain

$$E(\mathbf{Q}) \leq \frac{m_0}{m} q^* \leq q^*,$$

and the FDR is controlled.

Remark. Note that the independence of the test statistics corresponding to the false null hypotheses is not needed for the proof of the theorem.

This procedure was mentioned by Simes (1986) as an exploratory extension to his procedure for rejection of the intersection hypothesis that all null hypotheses are true if, for some i , $P_{(i)} \leq i\alpha/m$. Whereas Simes (1986) showed that his procedure controls the FWER under the intersection null hypothesis, Hommel (1988) showed that the extended procedure for inference on individual hypotheses does not control the FWER in the strong sense: for some configuration of the false null hypotheses, the probability of an erroneous rejection is greater than α .

Hochberg (1988) has suggested a different way to utilize Simes's procedure so that it does control the FWER in the strong sense, by offering the following procedure:

let k be the largest i for which $P_{(i)} \leq \frac{i}{m+1-i} \alpha$;

then reject all $H_{(i)}$ $i = 1, 2, \dots, k$.

Note the relationship between Hochberg's procedure and the FDR controlling procedure when q^* is chosen to equal α . Both Hochberg's procedure and the FDR controlling procedure are step-down procedures, which start by comparing $p_{(m)}$ with α , and if smaller all hypotheses are rejected—as if a PCER approach had been taken. If $p_{(m)} > \alpha$, proceed to smaller p -values until one satisfies the condition. The procedures end, if not terminated earlier, by comparing $p_{(1)}$ with α/m , as in a pure Bonferroni comparison. At the two ends the procedures are similar, but, in between, the sequence of $p_{(i)}$ s is compared with $\{1 - (i - 1)/m\}\alpha$ in the current procedure, rather than with $\{1/(m + 1 - i)\}\alpha$ in Hochberg's procedure. The series of linearly decreasing constants of the FDR controlling method is always larger than the hyperbolically decreasing constants of Hochberg, and the extreme ratio is as large as $4m/(m + 1)^2$ at $i = (m + 1)/2$. This shows that the suggested procedure rejects samplewise at least as many hypotheses as Hochberg's method and therefore has also greater power than other FWER controlling methods such as Holm's (1979).

3.2. Example of False Discovery Rate Controlling Procedure

Thrombolysis with recombinant tissue-type plasminogen activator (rt-PA) and anisoylated plasminogen streptokinase activator (APSAC) in myocardial infarction has been proved to reduce mortality. Neuhaus *et al.* (1992) investigated the effects of a new front-loaded administration of rt-PA *versus* those obtained with a standard regimen of APSAC, in a randomized multicentre trial in 421 patients with acute myocardial infarction. Four families of hypotheses can be identified in the study:

- (a) base-line comparisons (11 hypotheses), where the problem is of showing equivalence;
- (b) patency of infarct-related artery (eight hypotheses);
- (c) reocclusion rates of patent infarct-related artery (six hypotheses);
- (d) cardiac and other events after the start of thrombolytic treatment (15 hypotheses).

In this last family FDR control may be desired: we do not wish to conclude that the front-loaded treatment is better if it is merely equivalent to the previous treatment in all respects.

In the paper, however, there is no attention to the problem of multiplicity (the only exception being the division of the end points into primary and secondary). The individual p -values are reported as they are, with no word of warning regarding their interpretation. The authors conclude that

'Compared to APSAC treatment, despite more early reocclusions, the clinical course with rt-PA treatment is more favorable with fewer bleeding complications and a substantially lower in-hospital mortality rate, presumably due to improved early patency of the infarct-related artery'.

The statement about the mortality is based on a p -value of 0.0095.

Consider now the fourth family, which contains the comparison of mortality and 14 other comparisons. The ordered $p_{(i)}$ s for the 15 comparisons made are

$$0.0001, 0.0004, 0.0019, 0.0095, 0.0201, 0.0278, 0.0298, 0.0344, \\ 0.0459, 0.3240, 0.4262, 0.5719, 0.6528, 0.7590, 1.000.$$

Controlling the FWER at 0.05, the Bonferroni approach, using $0.05/15 = 0.0033$, rejects the three hypotheses corresponding to the smallest p -values. These hypotheses correspond to reduced allergic reaction, and to two different aspects of bleeding; they do not include the comparison of mortality. Using Hochberg's procedure leaves us with the same three hypotheses rejected. Thus the statement about a significant reduction in mortality is unjustified from the classical point of view.

Using the FDR controlling procedure with $q^* = 0.05$, we now compare sequentially each $p_{(i)}$ with $0.05i/15$, starting with $p_{(15)}$. The first p -value to satisfy the constraint is $p_{(4)}$ as

$$p_{(4)} = 0.0095 \leq \frac{4}{15} 0.05 = 0.013.$$

Thus we reject the four hypotheses having p -values which are less than or equal to 0.013. We may support now with appropriate confidence the statements about mortality decrease, of which we did not have sufficiently strong evidence before.

3.3. Another Look at False Discovery Rate Controlling Procedure

The above FDR controlling procedure can be viewed as a *post hoc* maximizing procedure, as the following theorem suggests.

Theorem 2. The FDR controlling procedure given by expression (1) is the solution of the following constrained maximization problem:

$$\begin{aligned} &\text{choose } \alpha \text{ that maximizes the number of rejections at this level, } r(\alpha), \\ &\text{subject to the constraint } \alpha m / r(\alpha) \leq q^*. \end{aligned} \quad (3)$$

Proof. Observe that, for each α , if $p_{(i)} \leq \alpha < p_{(i+1)}$, then $r(\alpha) = i$. Furthermore, as the ratio on the left-hand side of constraint (3) increases in α over the range on which $r(\alpha)$ is constant, it is enough to investigate α s which are equal to one of the $p_{(i)}$ s. This $\alpha = p_{(k)}$ satisfies the constraint because $\alpha / r(\alpha) = p_{(k)} / k \leq q^* / m$. By considering the largest potential α s (largest $p_{(i)}$ s) first, the procedure yields the α with the largest $r(\alpha)$ satisfying the constraint.

Thus the procedure has also the appearance of a simultaneous maximization of R and FDR control being attempted after experimentation. When each hypothesis is tested individually at level α , the expected number of wrong rejections satisfies $E(V) \leq \alpha m$. So, after observing the outcome of the experiment, an upper bound estimate of Q_e is $\alpha m / r(\alpha)$. In view of the observed p -values, the level α can now be chosen, by maximizing the observed number of rejections $r(\alpha)$ subject to the constraint on the implied FDR-like bound. As noted in the examples of Section 2.2 this aspect of the procedure is desirable.